

ADAPTIVE CONTEXT MEMORY ENGINE

ACME

A domain-neutral, replayable execution system for
trustworthy AI workflows

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The risk starts when one service owns everything

Coupled by default

Prompt construction, provider calls, domain interpretation, memory, state mutation and persistence often collapse into one service layer.

A model response can drift from candidate evidence into canonical state without an explicit trust boundary.

Domain changes, provider changes and recovery logic become entangled.

Consequences

Executions are difficult to audit or replay safely after failure.

Domain logic is hard to replace, and operational claims become difficult to prove.

ACME treats model output as a candidate, never as truth

The central design thesis

A probabilistic model can propose structured work. Deterministic contracts, domain policy and recorded evidence decide whether anything becomes canonical.

The architecture is built around explicit transitions between those trust levels.



Validate

Parse, runtime-schema check and semantic validation reject malformed or input-inconsistent output.



Interpret

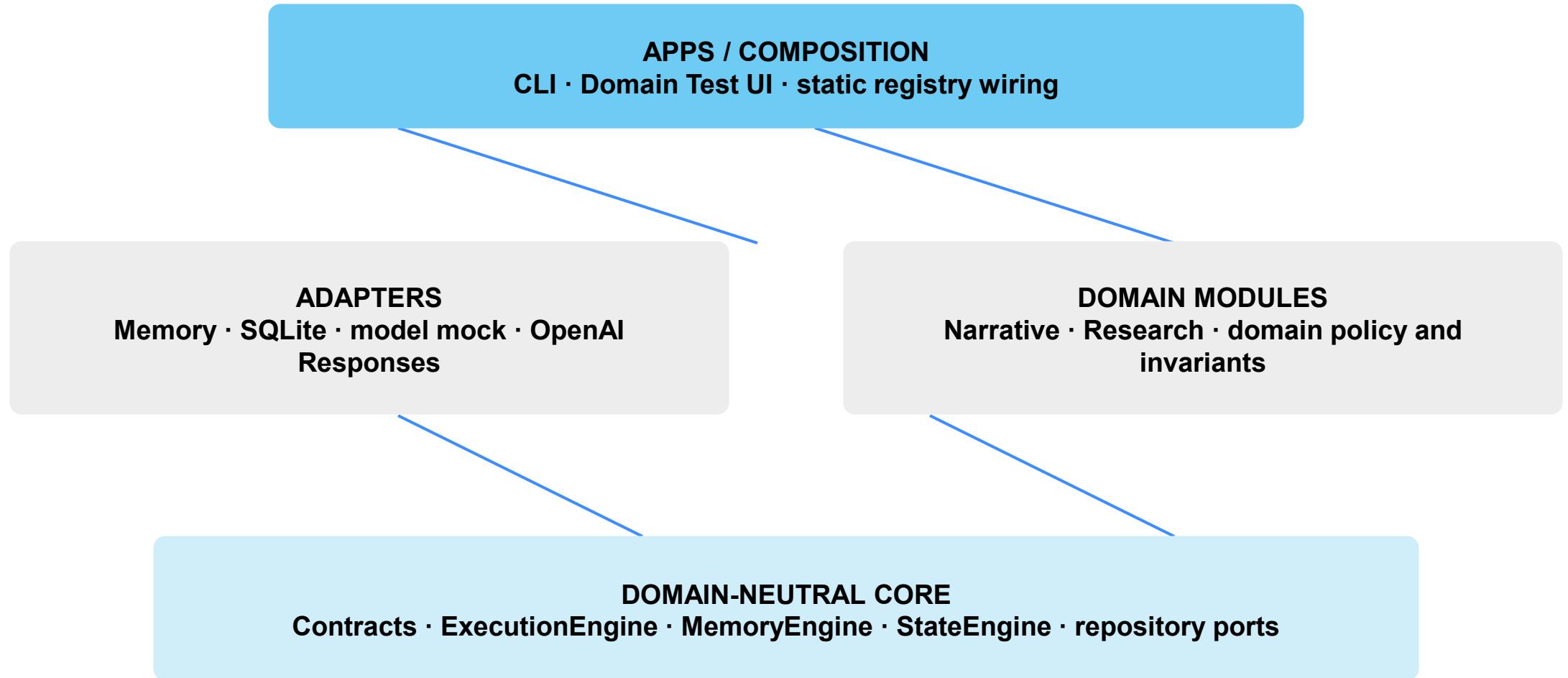
A task-typed domain module turns validated output into memory candidates and explicit state intent.



Commit

Memory policy, state reducer and invariants prepare one atomic, revision-checked Unit of Work.

The architecture makes dependency direction visible



Ports face inward. Provider SDKs, databases, transports and domain vocabulary stay outside core.

One execution crosses three explicit trust stages

1 · CANDIDATE

Contract gate

Input-bound request construction, provider call, parsing, schema and semantic validation.

2 · DOMAIN DECISION

Policy gate

Interpretation, memory resolution, post-memory projection, reducer and invariants.

3 · CANONICAL EVIDENCE

Atomic commit

Ledger, documents, memory, state and outbox commit together; replay recomputes the digest offline.

The workspace mirrors the architecture

A docs-first monorepo

The folder map is not cosmetic: package boundaries and public entry points mirror architectural ownership.

Negative dependency fixtures make forbidden shortcuts mechanically testable.

CORE

Contracts, deterministic identities, ExecutionEngine, MemoryEngine, StateEngine and repository ports.

MODULES

Narrative and Research own schemas, interpretation, memory policy, reducers and invariants.

ADAPTERS

In-memory, SQLite, deterministic model mock and OpenAI Responses mapping behind ports.

APPS + TESTING

CLI composition root, local Domain Test UI, reusable conformance kits and ScenarioRunner.

Two domains prove the core is not narrative-specific

NarrativeModule

Observes documents; preserves character facts, relationships and world rules in memory; projects scene, aliases and outline progress through a pure reducer.

ResearchModule

Observes evidence; tracks claims, sources, independence and contradictions; verifies only after corroboration and contests rather than overwrites.

Durability is behavior ACME proves, not assumes

Atomic commit

Documents, memory, state, events and the terminal result commit as one SQLite Unit of Work.

Safe concurrency

Revision compare-and-swap lets exactly one contended writer commit; the loser writes nothing.

Resume without re-calling

A recorded successful model call can complete after interruption with zero additional provider calls.

Replay + outbox

Replay recomputes the operation digest offline; committed events leave through an explicit at-least-once drain.

Quality is separated into different questions

ACME avoids one overloaded score. Exact assertions, population measurements, pre-commit gates, post-execution assessments and human fixture approval have different authority and lifecycle.

Each result remains attributable to the evidence and versioned contract that produced it.

A failed quality verdict is still a successful assessment; only an explicit assertion fails a scenario.

- ✓ Assertion · did an exact fact hold?
- ✓ Metric · how did a run population behave?
- ✓ Pre-commit gate · may the candidate commit?
- ✓ Post-exec quality · how good was one bound result?
- ✓ Fixture approval · may a proposed reference change?

ACME evolved through proof, not feature accumulation

Foundation

Docs-first chartering, typed contracts, pure state and memory engines, repositories, both domains and the bounded ExecutionEngine.

Live + durable

OpenAI Responses mapping, schema lowering, CLI, replay, encrypted retention, durable resume, rollback, CAS and outbox.

Workbench + quality

Domain Test UI S1–S10, S11 quality view, durable quality store, live-model judge and async launch with cancellation.

29–31 JUL

1 AUG

2–9 AUG

The delivered surface spans code, CLI and browser

Single execution

Typed task execution with idempotency and bounded model calls.

ScenarioRunner

Deterministic and opt-in live multi-step scenarios.

SQLite ledger

Durable evidence, state, memory and replay sidecars.

Outbox ops

Inspect, drain, redrive and file delivery.

Domain Test UI

Local S1–S10 workbench with progress and cancellation.

Quality

List, inspect and live-judge stored assessments.

Provider path

OpenAI Responses mapping behind an injected transport.

Conformance

Shared kits for repositories, gateways, modules and quality stores.

Evidence is exercised in independent test layers

Verification baseline · observed 2026-08-09

All default gates are offline and secret-free. In addition to the three counts below, 24 scenario tests in 5 files exercise end-to-end domain and quality flows.

603

unit tests
73 files

64

conformance tests
9 files

56

integration tests
10 files

Current status: a proven prototype, not a production platform

As of 2026-08-09

Milestones 1 and 2 are delivered. The experimental live path, complete local S1–S10 workbench, pure S11 quality view, durable quality store and async job path are implemented.

DELIVERED

Domain-neutral core + two domain proofs
Durable SQLite, replay, resume and outbox
CLI, local Test UI and quality evaluation

NOT CLAIMED

No published package or deployment
No production hosting or database decision
Live provider path remains experimental and opt-in

The remaining work is intentionally bounded

Next decisions, not hidden TODOs

Open items are inventoried as evidence gaps, interface residuals or deliberately deferred product choices. Each requires a new frozen charter before implementation.

Finer trust evidence

Record the exact failing substage inside memory, projection and state preparation instead of reporting only 'reached'.

UI residuals

Optional plan measurements, adapter declaration policy and browser CI smoke remain separate choices.

Provider + security

Ambiguous-call reconciliation, KMS/key rotation and privacy deletion are deferred behind evidence and ADRs.

Productization

Production hosting/database, dynamic discovery, vector retrieval and a workflow runtime beyond ScenarioRunner are not yet selected.

ACME turns model behavior into inspectable system behavior.

Translate domain rules into typed contracts and policies.

Integrate providers and persistence behind testable ports.

Prove failure modes before scaling the surface.